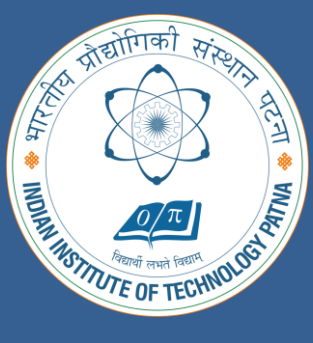




PRISMA: Preference-Reinforced Self-Training Approach for Interpretable Emotionally Intelligent Negotiation Dialogues

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Problem Statement

Develop an interpretable, emotionally intelligent negotiation dialogue system that perceives, understands, uses, and manages users' emotions to generate strategically appropriate, transparent, and emotionally attuned responses for job interview and resource allocation negotiations.

Motivation

Role of Emotions in Negotiation

Emotions drive concession-making, fairness, joint gains, and cooperation, making emotional intelligence essential for negotiation agents.

Need for Emotional Intelligence (EI)

EI (perceiving, understanding, using, managing emotions) is vital for high-stakes negotiations like job interviews and resource allocation.

Importance of Interpretability in Negotiation Agents

Beyond emotion-aware responses, users must understand how an agent generates them. Interpretability fosters reliability, builds rapport, and improves negotiation credibility.

Limitations of Existing LLM-based Negotiation Agents

LLMs generate static (not adaptive) emotion-aware strategies, struggle to distinguish genuine from manipulated emotions, and lack reasoning about emotional cause-and-effect dynamics.

Towards Interpretable, Emotionally Intelligent Negotiation Dialogue System

Motivated by these needs, PRISMA is designed to:

- Reason via ENS-CoT to perceive, understand, use, and manage user emotions
- Select appropriate emotion-aware strategies and ground responses in them
- Leverage a preference-reinforced self-training paradigm integrating supervised initialization, self-training, and DPO

Contribution

- Present PRISMA - an interpretable, emotionally intelligent negotiation dialogue agent, pioneering interpretability modeling in emotion-aware negotiation conversations;
- Propose an Emotion-aware Negotiation Strategy-informed Chain-of-Thought (ENS-CoT) reasoning mechanism that mimics human negotiation by perceiving, understanding, using, and managing emotions;
- Curate two novel datasets, JobNego (job interview negotiation) and ResNego (resource allocation negotiation), annotated with ENS-CoT rationales, and introduce a preference-reinforced self-training approach integrating self-training with DPO.
- Introduce a preference-reinforced self-training approach integrating self-training with DPO, demonstrating its effectiveness in producing interpretable, emotion-aware negotiation responses.

Dataset

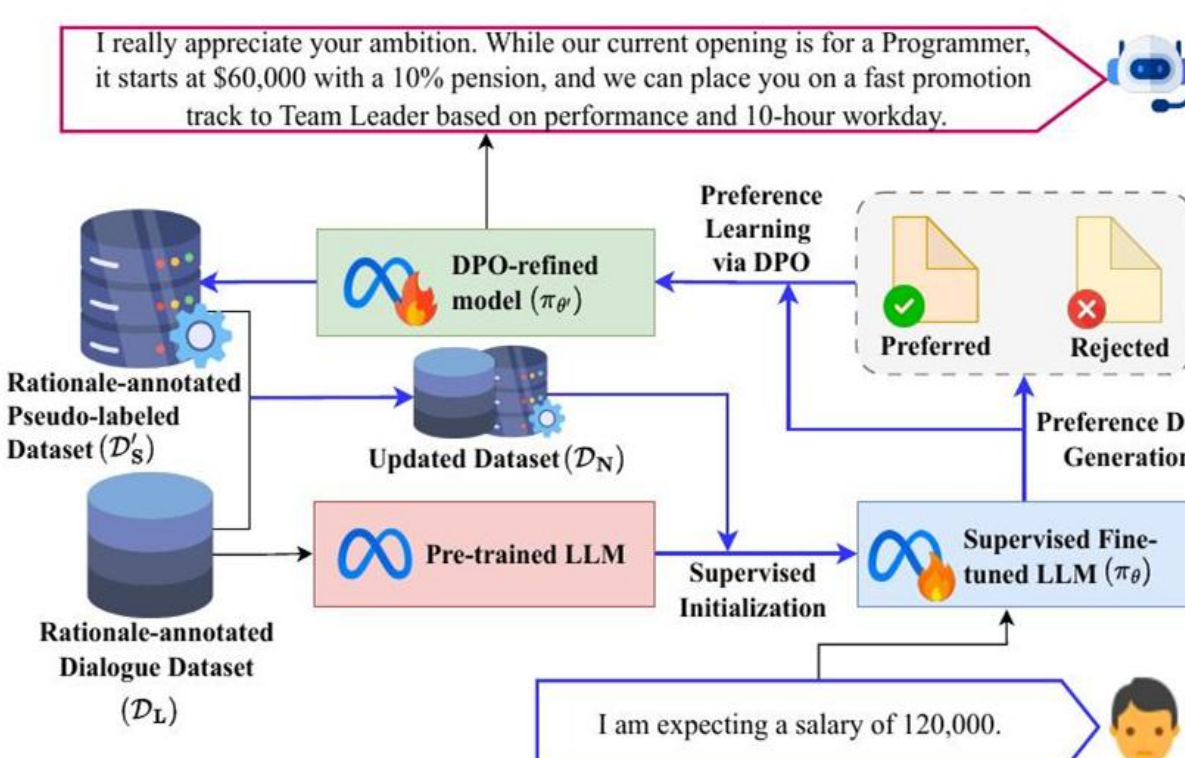
- JobNego Dataset**
 - 840 job interview negotiation dialogues between candidate (user) and employer (agent), covering salary, working hours, promotion track, and related terms.
- ResNego Dataset**
 - 1,648 resource-allocation negotiation dialogues in a camping setting, where parties bargain over food, water, firewood, and related supplies.
- Annotated with ENS-CoT rationales covering 12 emotions and 12 emotion-aware negotiation strategies via ChatGPT-based few-shot prompting with human-in-the-loop verification.

Annotation Aspects' Labels

Aspect	Label
Emotion (EM)	<i>Joy, Confidence, Positivity, Gratitude, Trust, Surprise, Anger, Disappointment, Frustration, Fear, Anxiety, Neutral</i>
Emotional Trigger (ET) & Individual Assessment (IA)	<i>Cause of emotion (external trigger or internal thought) and the user's personal interpretation, evaluation, and internal response to it, based on past experiences, beliefs, expectations, and values.</i>
Perspective Shift (PS) & Mindset Transformation (MT)	<i>Adopting an alternate viewpoint to reduce reactivity and trade-offs, and reframing limiting beliefs into constructive priors for decision-making.</i>
Strategy Selection (SS) & Strategy Reason (SR)	<i>Savoring, Positive Reinforcement, Expressing Optimism, Cognitive Reappraisal, Positive Framing, Emotion Diffusion, Expressive Suppression, Active Listening, Perspective-Taking, Problem Solving, Escalate Assurance, No Strategy</i>
Response Generation (RG)	<i>Using the selected emotion-aware negotiation strategy to generate the agent's emotionally attuned, interpretable, and strategically appropriate response in the ongoing dialogue.</i>

Proposed PRISMA System

- PRISMA comprises:
 - An Emotion-aware Negotiation Strategy-informed Chain-of-Thought (ENS-CoT) reasoning mechanism that mimics human negotiation by perceiving, understanding, using, and managing emotions
 - ENS-CoT octuple (EM, ET, IA, PS, MT, SS, SR, RG) - grounded in the MSCEIT's four abilities to enable interpretable emotional intelligence in negotiation responses
 - A Preference-Reinforced Self-Training paradigm integrating supervised initialization, self-training, and Direct Preference Optimization (DPO)
 - Training Procedure - *Step 1: Supervised Initialization on rationale data via. negative log-likelihood loss*, . *Step 2: Preference-Reinforced Self-Training alternating via. DPO-based preference learning and self-training with pseudo-labels.*



Evaluation Metrics

Language Quality Evaluation

- Automatic Evaluation**
 - Perplexity (PPL)**: How well a model predicts a response; **BLEU-4 (B-4)**: Overlap between ground-truth response and generated response; **BERTScore-F1 (BS-F1)**: Semantic similarity between gold response and generated response; **Distinct-3 (D-3)**: Diversity of the generated response; **Response Length (R-LEN)**: Length of the generated response
- Human Evaluation (Done on scale 1-5 by three human evaluators)**
 - F**: Assesses grammatical correctness and overall language quality; **C**: Measures whether responses are logically consistent and contextually relevant within the dialogue; **E**: Assesses how interesting and captivating the conversation is for the user

Goal Accomplishment Evaluation

- Automatic Evaluation and Human Evaluation (Done on scale 1-5 by three human evaluators)**
 - Emotion Appropriateness (EA)**: Evaluates the appropriateness of users' perceived emotions in the generated response; **Emotion-aware Negotiation Strategy Consistency (ENSC)**: Assesses the consistency of generated responses with the intended emotion-aware negotiation strategies; **Bargaining Efficacy (BE)**: Gauges the system's ability to bargain or present incentives that can influence the counterpart's decisions; **Outcome Fairness (OF)**: Examines whether the negotiation results in balanced and equitable agreements

Results and Analysis

Models	PPL↓	B-4↑	D-3↑	BS-F1↑	R-LEN↑	EA↑	ENSC↑
JobNego							
Llama-2-7B-SFT	19.87	3.68	29.92	0.633	13.36	0.479	0.646
Emotion-aware SFT	18.74	4.06	30.41	0.641	14.80	0.486	0.654
AnE	17.51	4.87	32.23	0.654	16.93	0.495	0.668
Standard	15.86	5.63	33.74	0.668	18.70	0.522	0.684
ProCoT (GPT-5)	19.21	3.51	29.75	0.629	17.55	0.471	0.639
RLHF	15.12	7.03	34.89	0.688	19.62	0.671	0.735
PRISMA	5.196	7.119	0.322	0.912	27.108	0.842	0.823
ResNego							
Llama-2-7B-SFT	3.76	3.66	35.88	0.762	19.15	0.358	0.393
Emotion-aware SFT	3.95	3.93	37.21	0.783	19.71	0.381	0.409
AnE	4.74	4.73	38.56	0.808	21.51	0.407	0.426
Standard	2.52	5.68	32.12	0.892	23.89	0.556	0.571
ProCoT (GPT-5)	9.84	3.46	34.78	0.812	27.81	0.504	0.533
RLHF	2.41	6.87	43.33	0.861	29.96	0.671	0.696
PRISMA	4.462	8.231	0.341	0.922	32.137	0.874	0.861

Table 1: Results of automatic evaluation.

Observations:

- PRISMA excels in negotiation dialogue quality, achieving superior fluency (low PPL), lexical overlap (B-4), semantic richness (BS-F1), diversity (D-3), and response length (R-LEN) across both JobNego and ResNego datasets.
- PRISMA attains the highest emotion appropriateness (EA) and emotion-aware negotiation strategy consistency (ENSC), validating the effectiveness of ENS-CoT rationales and preference-reinforced self-training in generating interpretable, emotion-aware negotiation responses.

Models	F↑	C↑	E↑	EA↑	ENSC↑	BE↑	OF↑
JobNego							
Llama-2-7B-SFT	2.94	2.87	2.76	2.32	2.28	2.13	3.28
Emotion-aware SFT	3.21	3.00	3.09	2.89	2.74	2.71	3.90
AnE	3.43	3.22	3.38	3.14	3.07	2.93	3.20
Standard	3.05	2.97	2.88	2.51	2.48	2.33	3.35
ProCoT (GPT-5)	3.68	3.54	3.51	3.45	3.42	3.30	3.60
RLHF	3.91	3.87	3.80	3.82	3.64	3.51	4.12
PRISMA	4.52	4.40	4.48	4.55	4.48	4.36	4.30
ResNego							
Llama-2-7B-SFT	2.85	2.73	2.52	2.48	2.61	2.37	3.45
Emotion-aware SFT	3.10	3.04	3.01	2.91	2.85	2.68	4.00
AnE	3.41	3.36	3.35	3.18	3.04	3.01	3.40
Standard	2.98	2.90	2.84	2.66	2.70	2.55	3.55
ProCoT (GPT-5)	2.62	3.59	3.44	3.39	3.36	3.28	3.80
RLHF	3.86	3.83	3.75	3.84	3.72	3.55	4.24
PRISMA	4.68	4.56	4.62	4.70	4.64	4.50	4.45

Table 2: Results of human evaluation

Observations:

- High EA and ENSC scores demonstrate PRISMA's effectiveness in maintaining emotional alignment and strategic consistency, while superior BE and OF reflect balanced, outcome-focused negotiation. Strong F, C, and E scores confirm fluent, coherent, and engaging dialogues, validating the interpretable ENS-CoT reasoning.

Conclusion

- Proposed PRISMA, an interpretable, emotionally intelligent negotiation dialogue agent for job interview and resource allocation negotiations.
- Proposed an Emotion-aware Negotiation Strategy-informed Chain-of-Thought (ENS-CoT) reasoning mechanism that links perceived emotions to strategy selection and response generation.
- Curated JobNego and ResNego datasets using ENS-CoT, and integrated self-training with Direct Preference Optimization (DPO) through a preference-reinforced self-training framework.
- Demonstrated through extensive automatic and human evaluation that PRISMA achieves superior emotion appropriateness, strategy consistency, and negotiation effectiveness with improved interpretability.

Dataset & Code

Contact



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